Using Eclipse
(just write code in eclipse all commands to run in terminal)
Eclipse : new -> java project
Make new folder (say src1)
In that folder: new -> file →(3 files)

In eclipse new java project → keep same java version → disable module (ie create module-info.java file)->finish→create new folder src1

In same project in src1 folder new→file →paste whole code with .java (for all 3 files)
No need to explicitly mention path same commands!!!

1. WordCount.java

```
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
      public class WordCount {
         public static void main(String[] args) throws Exception {
             Configuration conf = new Configuration();
             Job job = Job.getInstance(conf, "word count");
             job.setJarByClass(WordCount.class);
             job.setMapperClass(WordMapper.class);
             job.setCombinerClass(IntSumReducer.class);
             job.setReducerClass(IntSumReducer.class);
             job.setOutputKeyClass(Text.class);
             job.setOutputValueClass(IntWritable.class);
             FileInputFormat.addInputPath(job, new Path("input.txt"));
             FileOutputFormat.setOutputPath(job, new Path("output"));
             System.exit(job.waitForCompletion(true) ? 0 : 1);
         }
      }
```

2. WordMapper.java

```
import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;
public class WordMapper extends Mapper<Object, Text, Text, IntWritable> {
   private final static IntWritable one = new IntWritable(1);
   private Text word = new Text();
```

```
   public void map(Object key, Text value, Context context) throws IOException,
InterruptedException {
        String[] tokens = value.toString().split("\\s+");
        for (String token : tokens) {
            word.set(token.toLowerCase().replaceAll("[^a-zA-Z]", ""));
            if (!word.toString().isEmpty()) {
                context.write(word, one);
            }
        }
    }
}
```

3. IntSumReducer.java

```
import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;
public class IntSumReducer extends Reducer<Text, IntWritable, Text,
IntWritable> {
   public void reduce(Text key, Iterable<IntWritable> values, Context context)
throws IOException, InterruptedException {
        int sum = 0;
        for (IntWritable val : values) {
            sum += val.get();
        }
        context.write(key, new IntWritable(sum));
    }
}
```

Save all files
In that project create a untitled file (save as input.txt)

Go to terminal:

hadoop1@aakanksha:~/try$ start-all.sh
(switch on all nodes)

WARNING: Attempting to start all Apache Hadoop daemons as hadoop1 in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [aakanksha]
Starting resourcemanager

Starting nodemanagers

hadoop1@aakanksha:~$ cd try
Here try is my folder name where java project is created

hadoop1@aakanksha:~/try$ ls
bin input.txt src src1

(Just verifying that all things are there
  src1 is folder where all 3 java files are saved)

hadoop1@aakanksha:~/try$ mkdir -p classes
(Creating a folder classes to store compiled .java files if not created then it gets automatically created)

hadoop1@aakanksha:~/try$ ls
bin classes input.txt src src1

hadoop1@aakanksha:~/try$ javac -classpath "$(hadoop classpath)" -d classses src1/*.java
(to create .class files of all 3 java files
Here I have done error it should be classes not **classses**
)

hadoop1@aakanksha:~/try$ cd classses

hadoop1@aakanksha:~/try/classses$ jar -cvf wordcount.jar *.class
added manifest
adding: IntSumReducer.class(in = 1593) (out= 663)(deflated 58%)
adding: WordCount.class(in = 1412) (out= 780)(deflated 44%)
adding: WordMapper.class(in = 1945) (out= 862)(deflated 55%)
(creating jar file)

hadoop1@aakanksha:~/try/classses$ ls
IntSumReducer.class WordCount.class **wordcount.jar** WordMapper.class

hadoop1@aakanksha:~/try/classses$ cd ..

hadoop1@aakanksha:~/try$ hadoop fs -mkdir -p /user/hadoop1

hadoop1@aakanksha:~/try$ hadoop fs -put input.txt /user/hadoop1
(hadoop1 is user name)

hadoop1@aakanksha:~/try$ hadoop jar classses/wordcount.jar WordCount input.txt

(output named file gets created)

hadoop jar classes/wordcount.jar WordCount input.txt output

```
2025-04-25 18:03:34,079 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /127.0.0.1:8032
2025-04-25 18:03:34,310 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2025-04-25 18:03:34,329 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/hadoop1/.staging/job_1745584084518_0001
2025-04-25 18:03:34,494 INFO input.FileInputFormat: Total input files to process : 1
2025-04-25 18:03:34,551 INFO mapreduce.JobSubmitter: number of splits:1
2025-04-25 18:03:34,661 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1745584084518_0001
2025-04-25 18:03:34,661 INFO mapreduce.JobSubmitter: Executing with tokens: []
2025-04-25 18:03:34,794 INFO conf.Configuration: resource-types.xml not found
2025-04-25 18:03:34,794 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2025-04-25 18:03:35,178 INFO impl.YarnClientImpl: Submitted application application_1745584084518_0001
2025-04-25 18:03:35,202 INFO mapreduce.Job: The url to track the job: http://aakanksha:8088/proxy/application_1745584084518_0001/
2025-04-25 18:03:35,202 INFO mapreduce.Job: Running job: job_1745584084518_0001
2025-04-25 18:03:41,288 INFO mapreduce.Job: Job job_1745584084518_0001 running in uber mode : false
2025-04-25 18:03:41,290 INFO mapreduce.Job:  map 0% reduce 0%
2025-04-25 18:03:44,327 INFO mapreduce.Job:  map 100% reduce 0%
2025-04-25 18:03:48,359 INFO mapreduce.Job:  map 100% reduce 100%
2025-04-25 18:03:49,384 INFO mapreduce.Job: Job job_1745584084518_0001 completed successfully
2025-04-25 18:03:49,435 INFO mapreduce.Job: Counters: 54
        File System Counters
                FILE: Number of bytes read=115
                FILE: Number of bytes written=617517
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=182
                HDFS: Number of bytes written=69
                HDFS: Number of read operations=8
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
                HDFS: Number of bytes read erasure-coded=0
        Job Counters
```

```
        Launched map tasks=1
        Launched reduce tasks=1
        Data-local map tasks=1
        Total time spent by all maps in occupied slots (ms)=1491
        Total time spent by all reduces in occupied slots (ms)=1698
        Total time spent by all map tasks (ms)=1491
        Total time spent by all reduce tasks (ms)=1698
        Total vcore-milliseconds taken by all map tasks=1491
        Total vcore-milliseconds taken by all reduce tasks=1698
        Total megabyte-milliseconds taken by all map tasks=1526784
        Total megabyte-milliseconds taken by all reduce tasks=1738752
Map-Reduce Framework
        Map input records=4
        Map output records=14
        Map output bytes=130
        Map output materialized bytes=115
        Input split bytes=109
        Combine input records=14
        Combine output records=10
        Reduce input groups=10
        Reduce shuffle bytes=115
        Reduce input records=10
        Reduce output records=10
        Spilled Records=20
        Shuffled Maps =1
        Failed Shuffles=0
        Merged Map outputs=1
        GC time elapsed (ms)=22
        CPU time spent (ms)=1050
        Physical memory (bytes) snapshot=599814144
        Virtual memory (bytes) snapshot=5491290112
        Total committed heap usage (bytes)=520093696
        Peak Map Physical memory (bytes)=337338368
        Peak Map Virtual memory (bytes)=2740539392
        Peak Reduce Physical memory (bytes)=262475776
        Peak Reduce Virtual memory (bytes)=2752118784
Shuffle Errors
        BAD_ID=0
        CONNECTION=0
        IO_ERROR=0
        WRONG_LENGTH=0
        WRONG_MAP=0
        WRONG_REDUCE=0
File Input Format Counters
```

```
		Bytes Read=73
	File Output Format Counters
		Bytes Written=69
```

hadoop1@aakanksha:~/try$ hadoop fs -cat /user/hadoop1/output/part-r-00000
(print output from output file)

```
count	1
hadoop		3
hello	2
is	1
of	1
prof	1
run	1
test	1
this	2
word	1
```

When re executing the file input.txt and output files already exists so:

hadoop1@aakanksha:~/try$ hadoop fs -rm -r /user/hadoop1/input.txt
Deleted /user/hadoop1/input.txt

hadoop1@aakanksha:~/try$ hadoop fs -rm -r /user/hadoop1/output
Deleted /user/hadoop1/output

hadoop1@aakanksha:~/try$ hadoop fs -ls /user/hadoop1

(deleted both files and checking it)

hadoop1@aakanksha:~/try$ stop-all.sh
(switch off all nodes)

```
WARNING: Stopping all Apache Hadoop daemons as hadoop1 in 10 seconds.
WARNING: Use CTRL-C to abort.
Stopping namenodes on [localhost]
Stopping datanodes
Stopping secondary namenodes [aakanksha]
Stopping nodemanagers
localhost: WARNING: nodemanager did not stop gracefully after 5 seconds: Trying to kill with kill
-9
Stopping resourcemanager
```

SCALA

Easier one

In text editor:

```
// Step 1: Read input file from /home/input.txt
val inputFile = "file:///home/aakanksha/SEM_6_Labs/DSBDA/input.txt" actual path to your input file
val input = sc.textFile(inputFile)

// Step 2: Perform word count
val words = input.flatMap(line => line.split("\\s+"))
val wordPairs = words.map(word => (word, 1))
val wordCounts = wordPairs.reduceByKey(_ + _)

// Step 3: Save result to /home/output
val outputDir = "file:///home/aakanksha/SEM_6_Labs/DSBDA/output" actual  path where u want output
wordCounts.coalesce(1).saveAsTextFile(outputDir)
```

(Save as any_name.scala)

Terminal:

aakanksha@aakanksha:~$ spark-shell
(start spark shell)

```
25/04/25 18:35:23 WARN Utils: Your hostname, aakanksha resolves to a loopback address: 127.0.1.1; using 192.168.1.4 instead (on interface wlp0s20f3)
25/04/25 18:35:23 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
Welcome to
```

```
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /___/ .__/\_,_/_/ /_/\_\   version 3.4.1
      /_/
```

Using Scala version 2.13.8 (OpenJDK 64-Bit Server VM, Java 11.0.27)
Type in expressions to have them evaluated.
Type :help for more information.
25/04/25 18:35:26 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Spark context Web UI available at http://192.168.1.4:4040
Spark context available as 'sc' (master = local[*], app id = local-1745586327837).
Spark session available as 'spark'.

scala> :load try.scala (any_name.scala)
val args: Array[String] = Array()
Loading try.scala...
val inputFile: String = file:///home/aakanksha/SEM_6_Labs/DSBDA/input.txt
val input: org.apache.spark.rdd.RDD[String] = file:///home/aakanksha/SEM_6_Labs/DSBDA/input.txt MapPartitionsRDD[1] at textFile at try.scala:1
val words: org.apache.spark.rdd.RDD[String] = MapPartitionsRDD[2] at flatMap at try.scala:1
val wordPairs: org.apache.spark.rdd.RDD[(String, Int)] = MapPartitionsRDD[3] at map at try.scala:1
val wordCounts: org.apache.spark.rdd.RDD[(String, Int)] = ShuffledRDD[4] at reduceByKey at try.scala:1
val outputDir: String = file:///home/aakanksha/SEM_6_Labs/DSBDA/output

scala> :quit